UNIVERSITY OF MIAMI
DEPARTMENT OF BIOMEDICAL ENGINEERING

BME 707 - MS PROJECT ABSTRACT

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Student Name: Giacomo Albergo

Project mentor: Abhishek Prasad, PhD, College of Engineering

Project coordinator: Fotios M Andreopoulos, PhD, College of Engineering

Project description (100 to 200 words):

Brain-Computer Interfaces (BCIs) based on Motor Imagery (MI) have shown significant promise in neurorehabilitation and assistive technology. However, achieving high decoding accuracy in multiclass scenarios remains a critical challenge, particularly when transitioning from binary (e.g., left vs. right hand) to complex functional tasks. This Master's project investigates the development of a robust deep learning classifier designed for the real-time decoding of nine distinct hand movements using Electroencephalography (EEG) signals.

The study implements a neural network pipeline within a Python-based environment, utilizing the BIDS standard for data organization. The research focuses on two primary objectives: (1) establishing a high-accuracy multiclass classification model by overcoming common overfitting issues observed in standard architectures, and (2) analyzing the signal relationship between non-invasive EEG and invasive Electrocorticography (ECoG) to inform feature extraction strategies.

Preliminary experiments involved the implementation of a deep learning model on a multiclass dataset (9-class) and a simplified binary subset (Finger vs. Wrist). While initial baseline models highlighted the difficulty of multiclass separation with limited data, this project proposes advanced preprocessing and regularization techniques to enhance generalization. The final objective is to optimize the decoder for real-time application, aiming to bridge the gap between theoretical deep learning capacity and practical, high-accuracy BCI control.

Approved:

Abhishek Prasad

Project Mentor

Project Coordinator